

ECE 3317

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Review 1

Complex Vectors & EM theories

- Complex vectors and vector calculus
- Time domain $\leftrightarrow$ phasor domain
- Maxwell's equations
- Poynting theorem

Time Domain $\leftrightarrow$ Phasor Domain

$$V(t) = V_0 \cos(\omega t + \phi)$$

$$\left\{ \begin{array}{ll} \mathbf{V} = V_0 e^{j\phi} & \text{going from time domain to phasor domain} \\ V(t) = \text{Re}\{\mathbf{V}e^{j\omega t}\} & \text{going from phasor domain to time domain} \end{array} \right.$$

Time-domain $\Leftrightarrow$ Phasor domain

$$V(t) \Leftrightarrow \mathbf{V}$$

For vectors:

$$\underline{V}(t) = \underline{\hat{x}}V_x \cos(\omega t + \phi_x) + \underline{\hat{y}}V_y \cos(\omega t + \phi_y) + \underline{\hat{z}}V_z \cos(\omega t + \phi_z)$$
$$\underline{\mathbf{V}} \equiv \underline{\hat{x}}V_x e^{j\phi_x} + \underline{\hat{y}}V_y e^{j\phi_y} + \underline{\hat{z}}V_z e^{j\phi_z}$$

Maxwell's Equations

in time domain

$$\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t}$$

Faraday's law

$$\nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t}$$

Ampere's law

$$\nabla \cdot \underline{B} = 0$$

Magnetic Gauss law

$$\nabla \cdot \underline{D} = \rho_v$$

Electric Gauss law

in phasor domain

$$\nabla \times \underline{E} = -j\omega \underline{B}$$

$$\nabla \times \underline{H} = \underline{J} + j\omega \underline{D}$$

$$\nabla \cdot \underline{B} = 0$$

$$\nabla \cdot \underline{D} = \rho_v$$

Poynting Theorem

$$-\oint_S (\underline{E} \times \underline{H}) \cdot \hat{n} dS = \int_V \sigma |\underline{E}|^2 dV + \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \mu |\underline{H}|^2 \right) dV + \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \varepsilon |\underline{E}|^2 \right) dV$$

Power dissipation as heat (Joule's law)

Rate of change of stored magnetic energy

Rate of change of stored electric energy

power flowing into the region V .

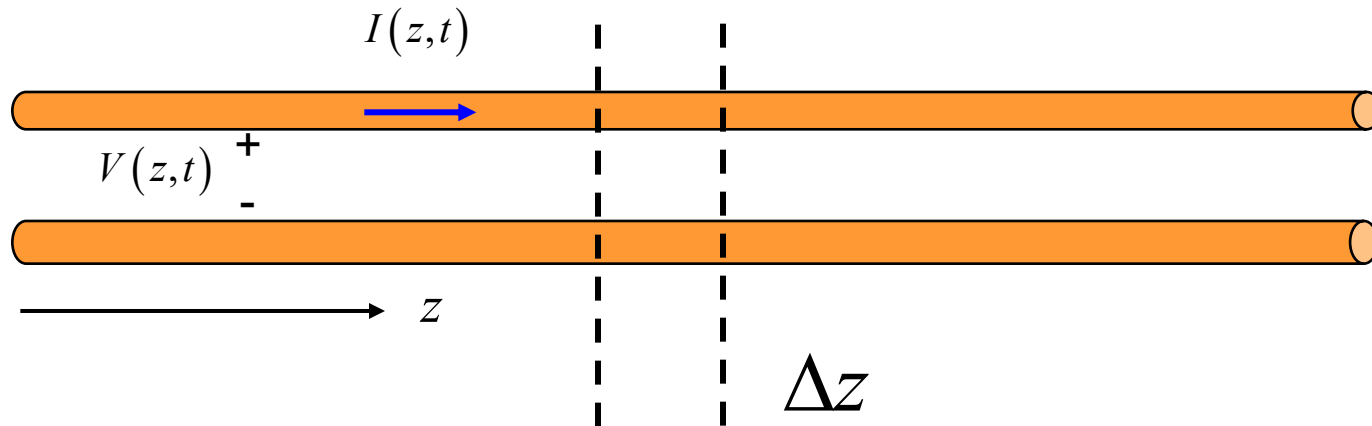
instantaneous Poynting vector: $\underline{S}(t) = \underline{E}(t) \times \underline{H}(t)$

instantaneous power flowing out of the region $= \oint_S \underline{S}(t) \cdot \hat{n} dS$

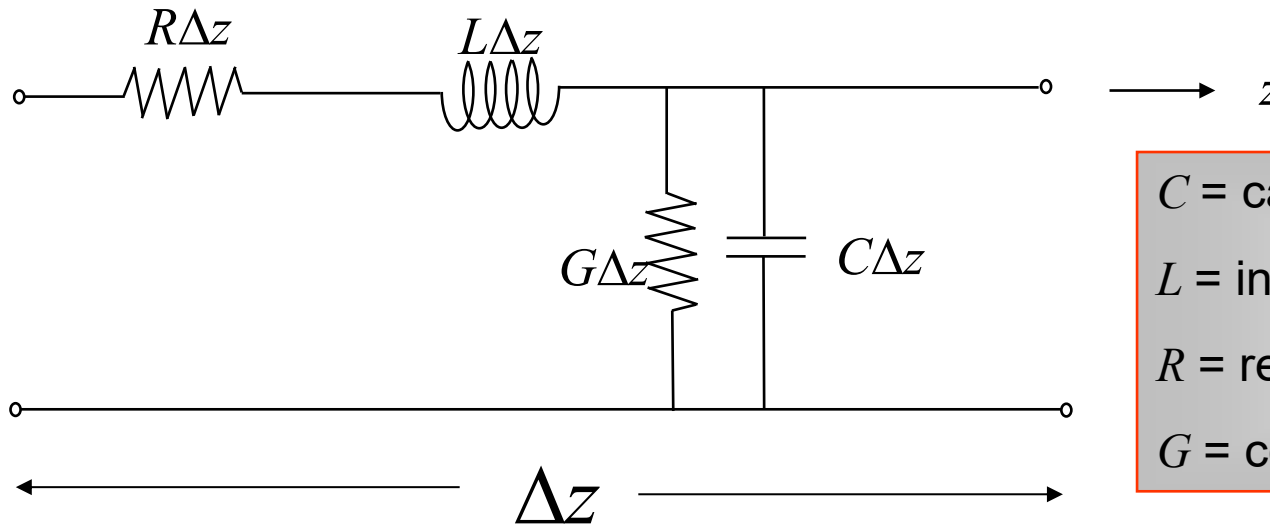
complex Poynting vector: $\underline{S} = \frac{1}{2} (\underline{E} \times \underline{H}^*)$

time-average power flowing out of the region $= \oint_S \text{Re} \{ \underline{S} \} \cdot \hat{n} dS$

Circuit Model of Transmission Line



Circuit Model:



C = capacitance/length [F/m]
 L = inductance/length [H/m]
 R = resistance/length [Ω /m]
 G = conductance/length [S/m]

Equations of Transmission Line

“Telegrapher’s Equations (TEs)”

$$\begin{aligned}\frac{\partial V}{\partial z} &= -RI - L \frac{\partial I}{\partial t} \\ \frac{\partial I}{\partial z} &= -GV - C \frac{\partial V}{\partial t}\end{aligned}$$

traveling wave solution

$$V(z, t) = f(z - c_d t) + g(z + c_d t)$$

where f and g are arbitrary functions.

$$c_d = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{LC}}$$

Speed of light in dielectric / transmission line

Lossy wave equation

(no exact solution)

$$\frac{\partial^2 V}{\partial z^2} - (RG)V - (RC + LG) \frac{\partial V}{\partial t} - LC \left(\frac{\partial^2 V}{\partial t^2} \right) = 0$$

Lossless case:

$$R = G = 0$$

“wave equation”

$$\frac{\partial^2 V}{\partial z^2} - LC \left(\frac{\partial^2 V}{\partial t^2} \right) = 0$$

$$\frac{\partial^2 V}{\partial z^2} - \frac{1}{c_d^2} \left(\frac{\partial^2 V}{\partial t^2} \right) = 0 \quad c_d = \frac{1}{\sqrt{LC}}$$

Traveling Wave

$$V(z, t) = f(z - c_d t)$$



forward traveling wave

$$V(z, t) = g(z + c_d t)$$



backward traveling wave

characteristic impedance Z_0

$$Z_0 = \sqrt{\frac{L}{C}}$$

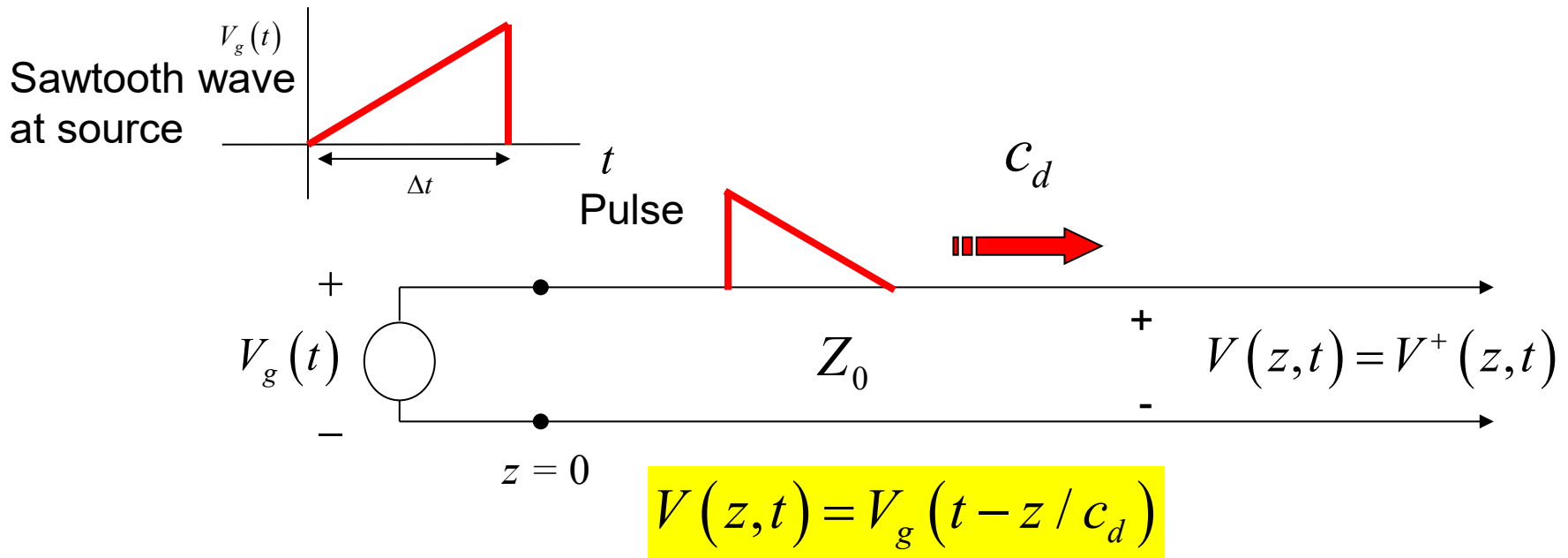
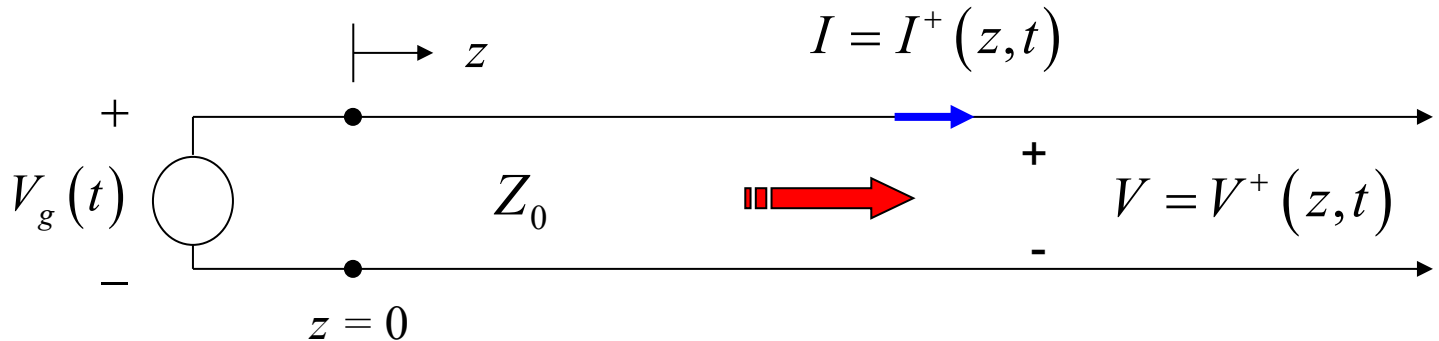
The units of Z_0 are Ohms.

$$V(z, t) = f(z - c_d t) + g(z + c_d t)$$

$$I(z, t) = \frac{1}{Z_0} [f(z - c_d t) - g(z + c_d t)]$$

Wave Propagation in Time Domain

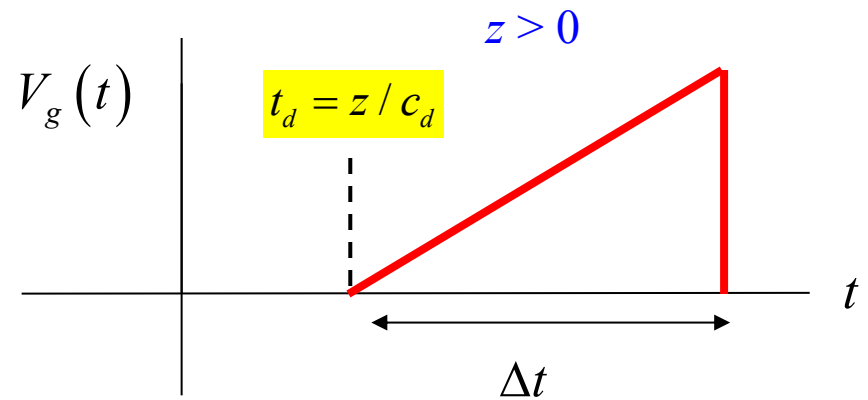
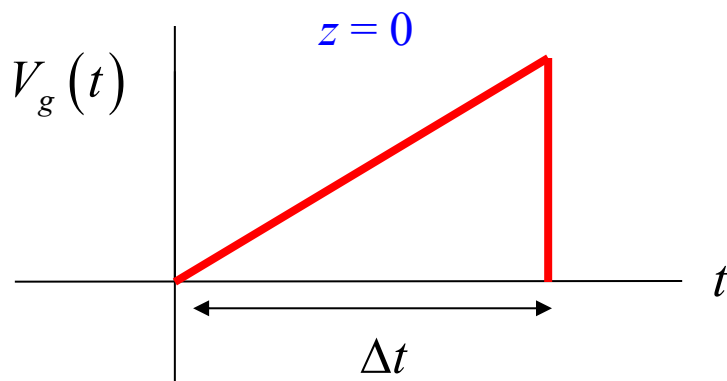
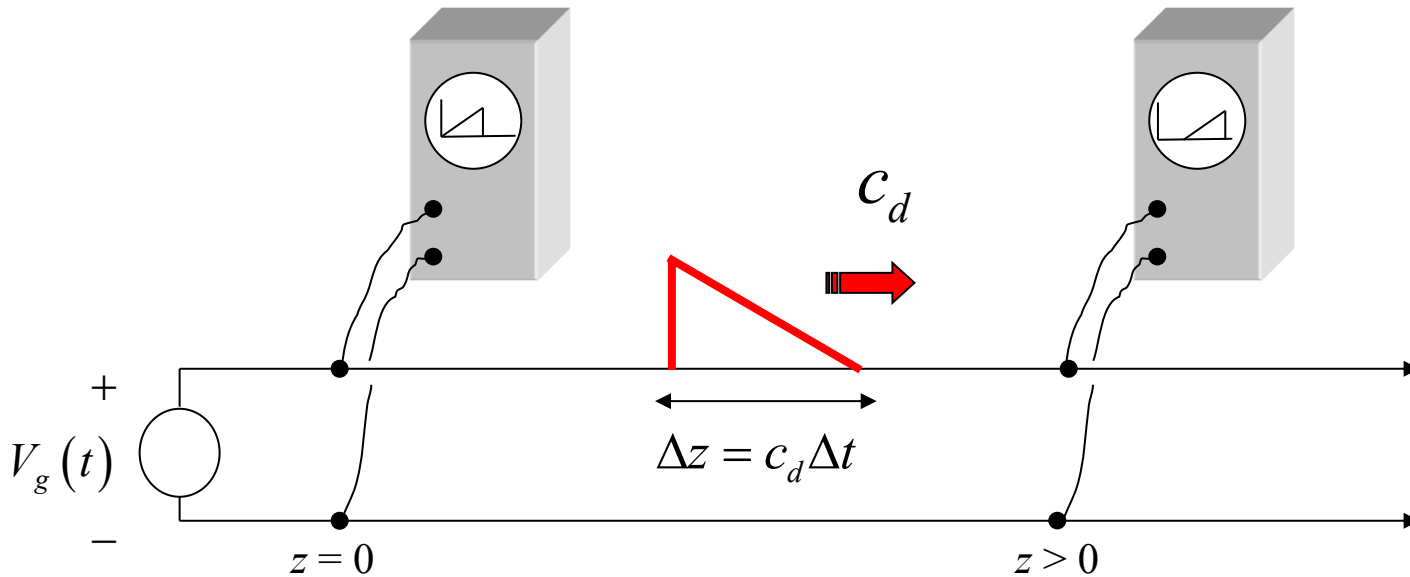
A voltage signal is applied at the input of the **semi-infinite** transmission line.



Note that the shape of the pulse as a function of z is a scaled mirror image of the pulse shape as a function of t .

Oscilloscope trace

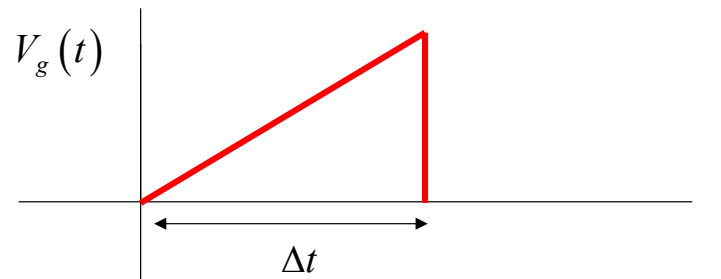
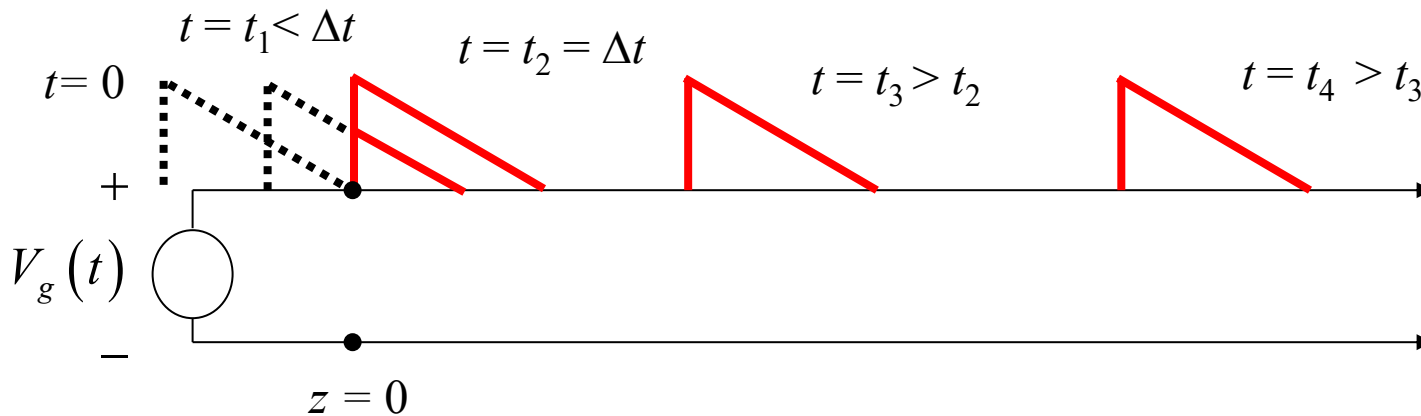
Note the delay in the trace on this oscilloscope.



Snapshots

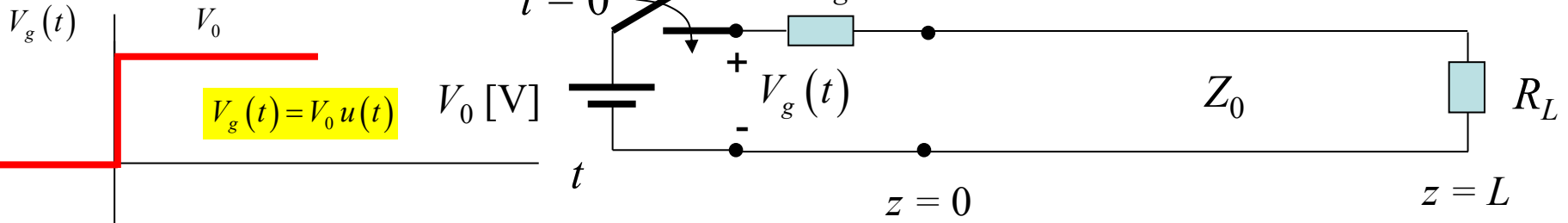
The pulse is shown emerging from the source end of the line.

A series of “snapshots” is shown.



Bounce Diagram

Bounce diagram is for a unit step response on a terminated line.

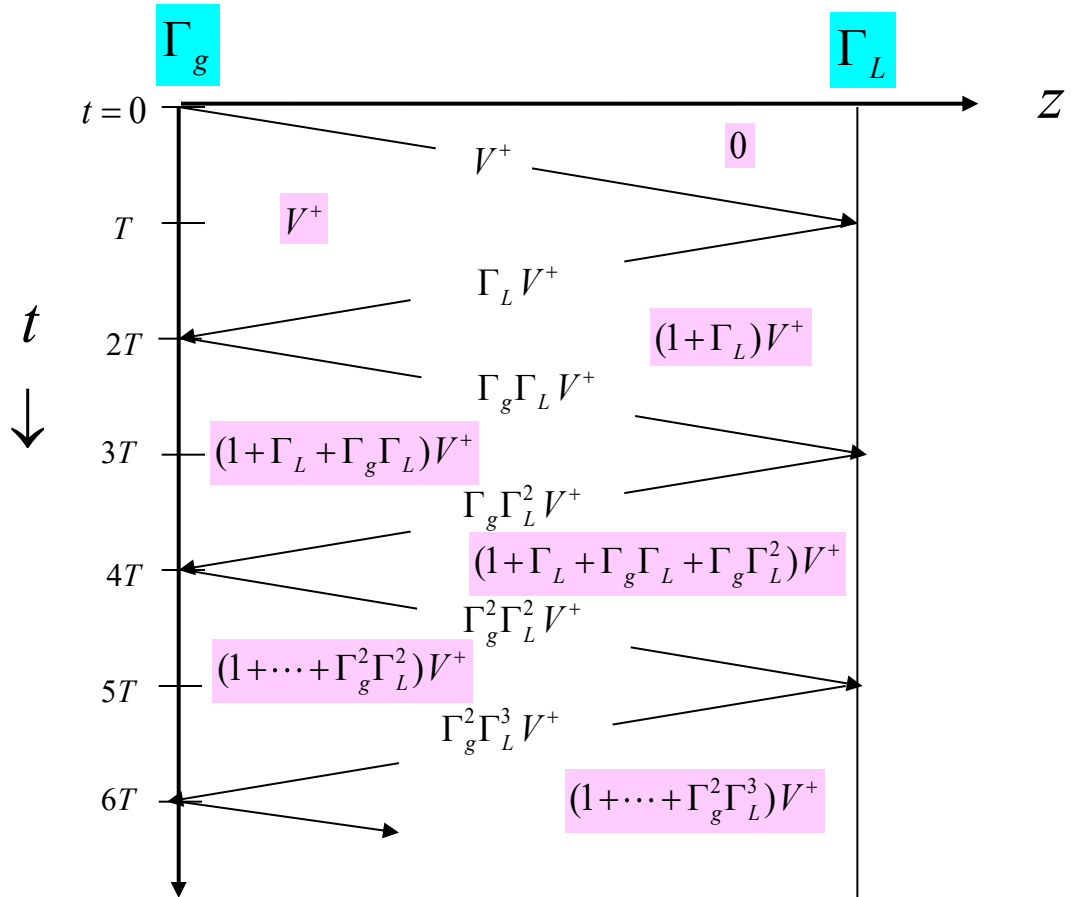


$$T = \frac{L}{c_d}$$

$$V^+ = \left(\frac{Z_0}{R_g + Z_0} \right) V_0$$

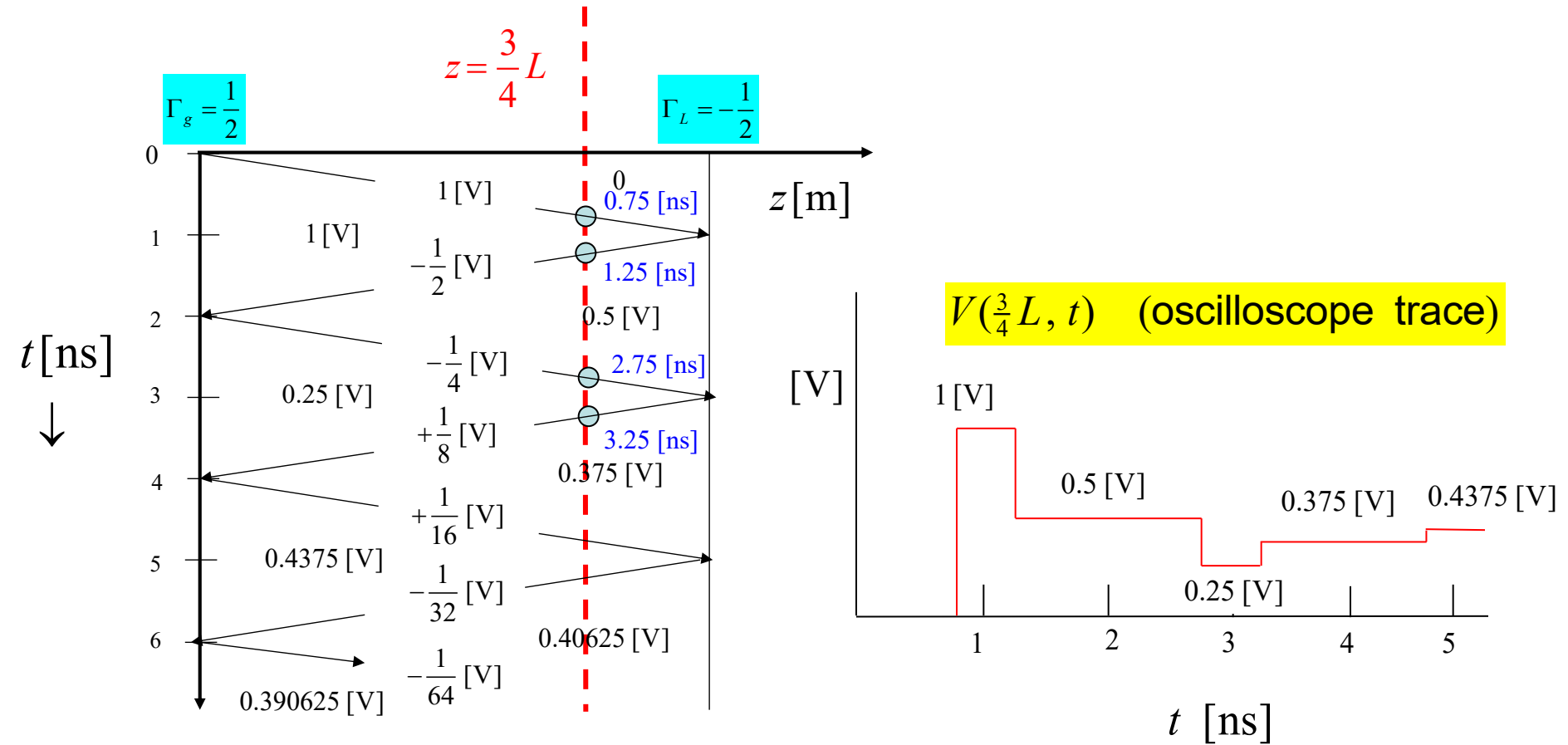
$$\Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right)$$



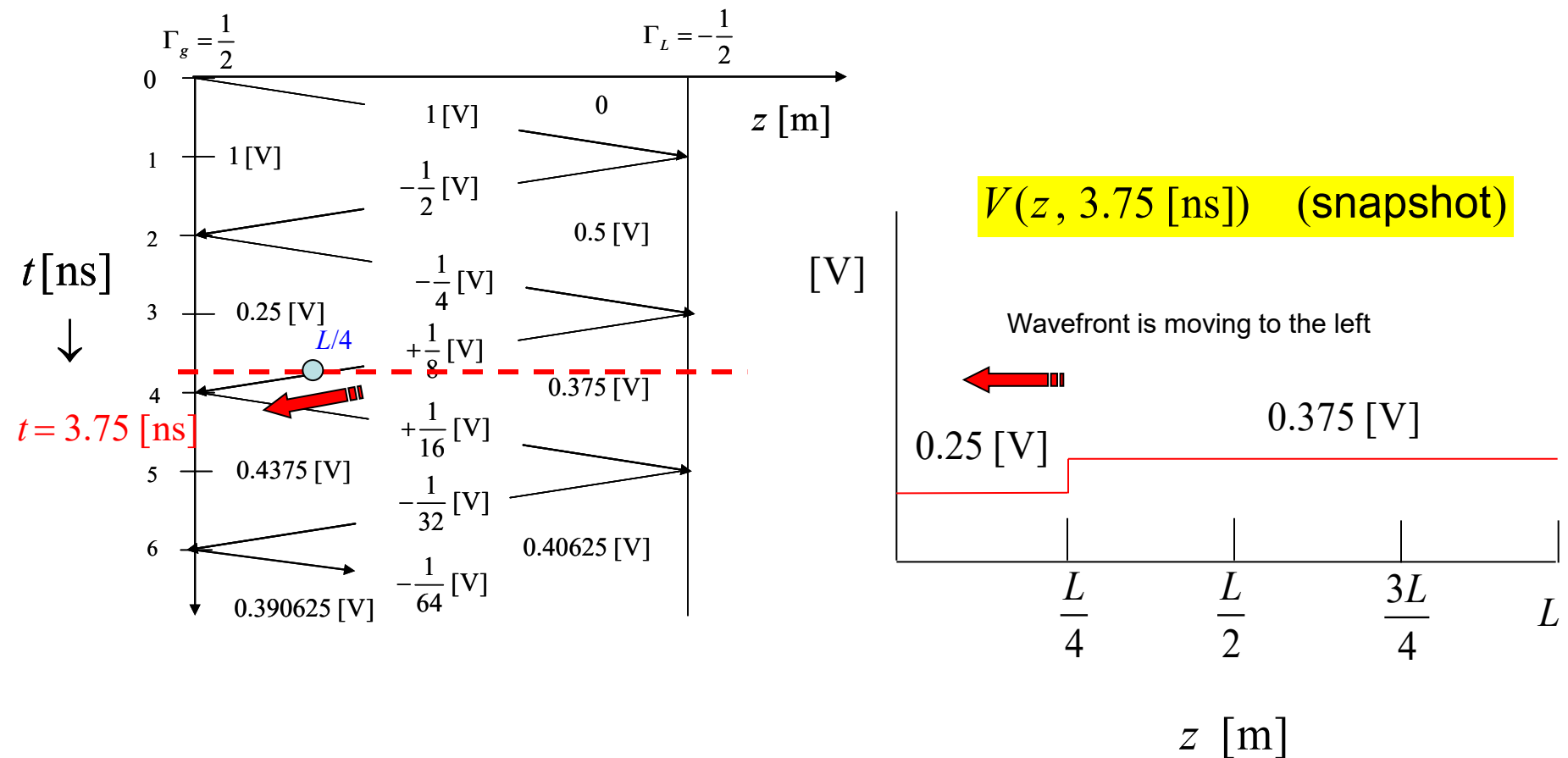
Oscilloscope Trance

Bounce diagram can be used to get an “oscilloscope trace” at any point on the line.



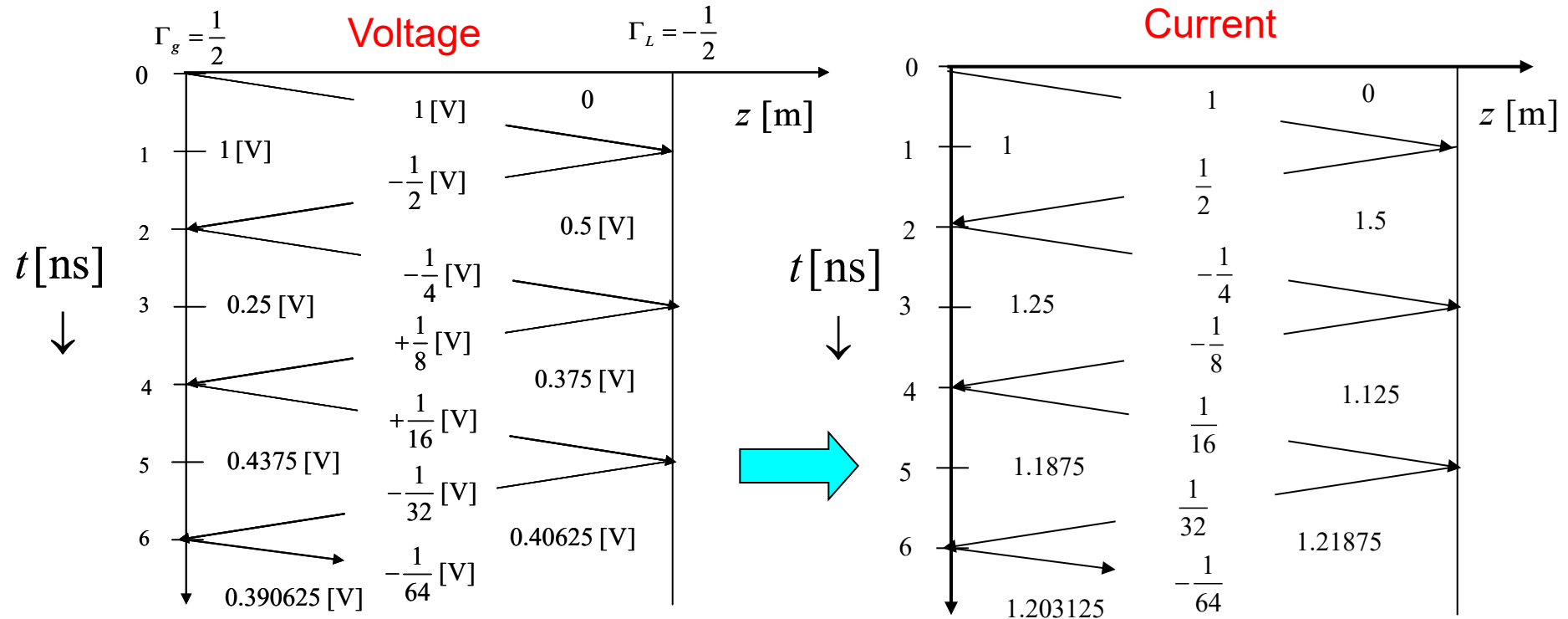
Snapshot

The bounce diagram can also be used to get a “snapshot” of the line voltage at any point in time.



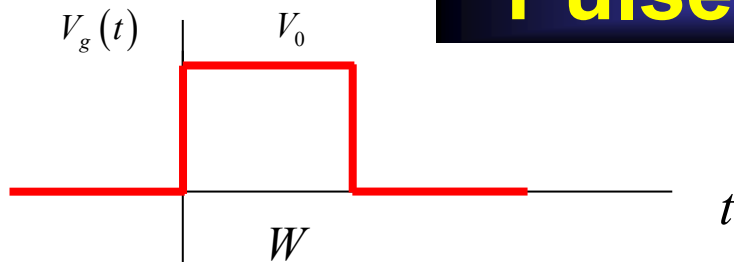
Bounce Diagram for Current

To obtain a **current bounce diagram** from the voltage diagram, multiply forward-traveling voltages by $1/Z_0$, backward-traveling voltages by $-1/Z_0$.



Note: This diagram is for the *normalized* current, defined as $Z_0 I(z, t)$.

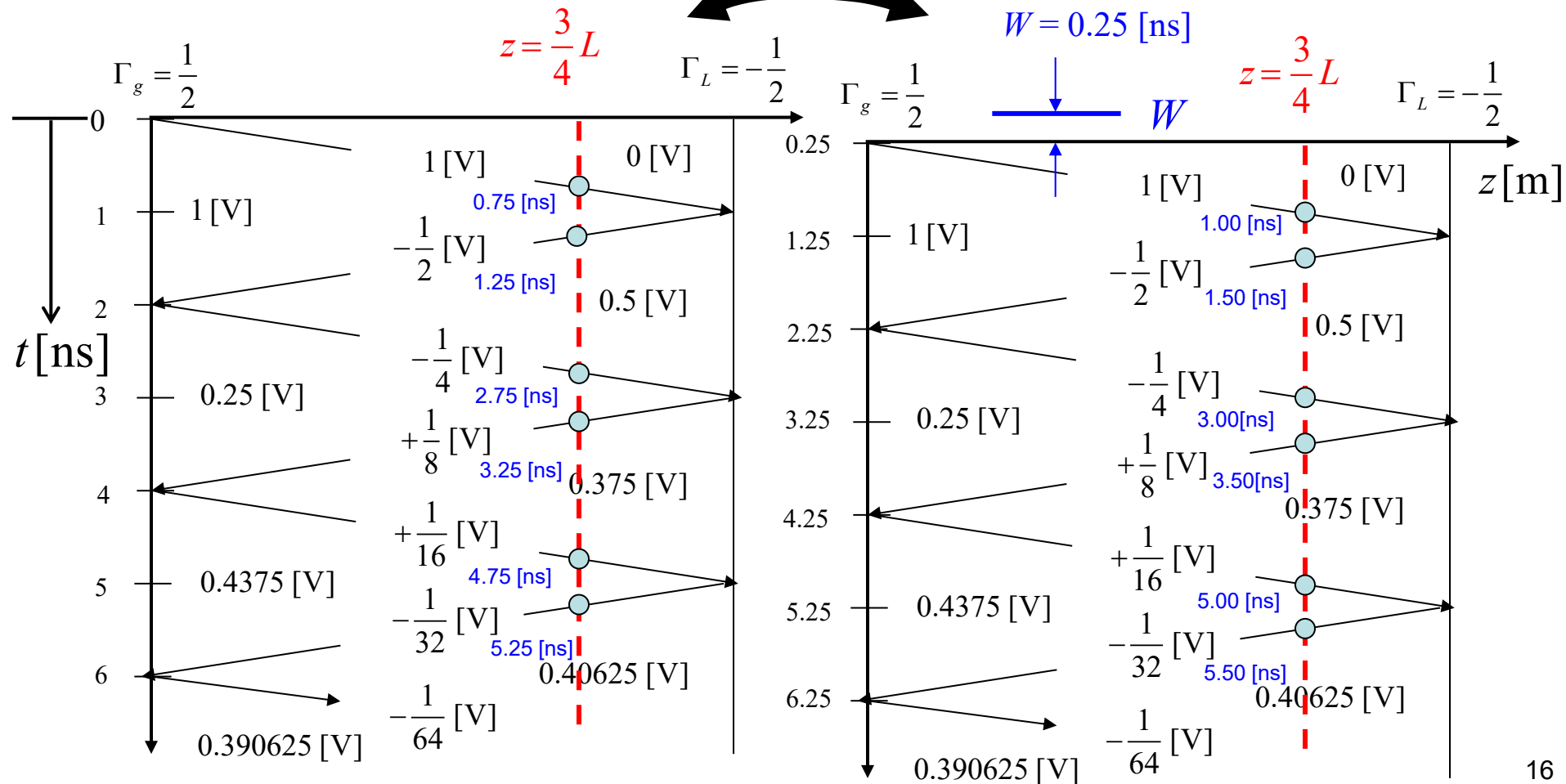
Pulse Response



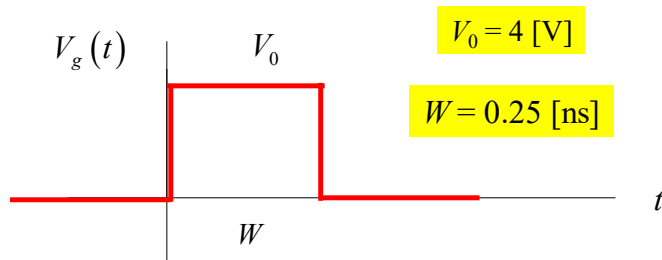
$$V_g(t) = V_0(u(t) - u(t - W))$$

Subtract

We subtract two bounce diagrams, with the second one being a shifted version of the first one.

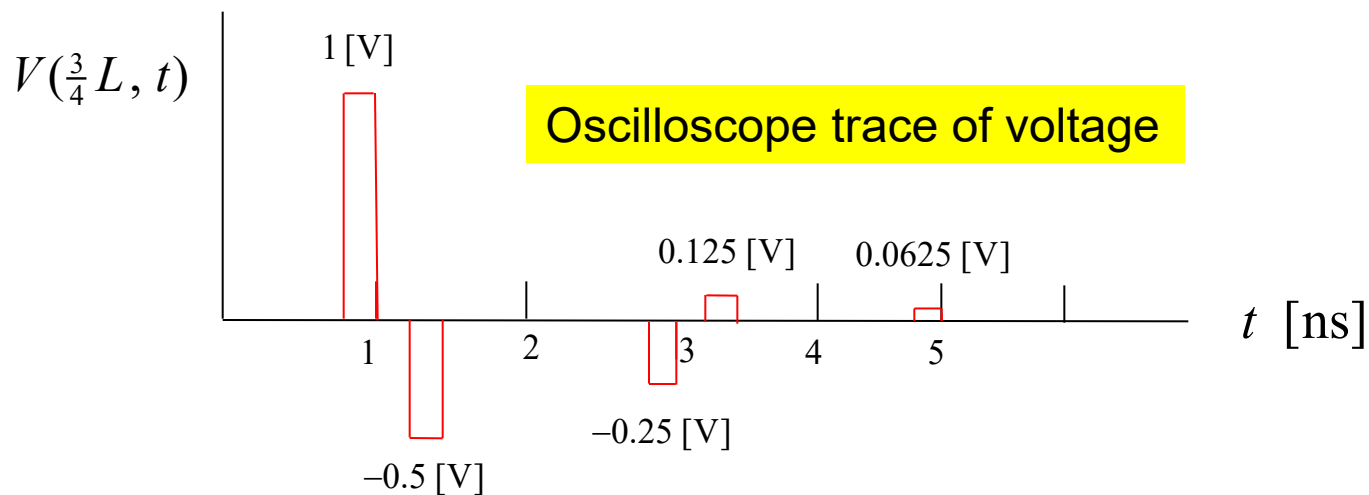
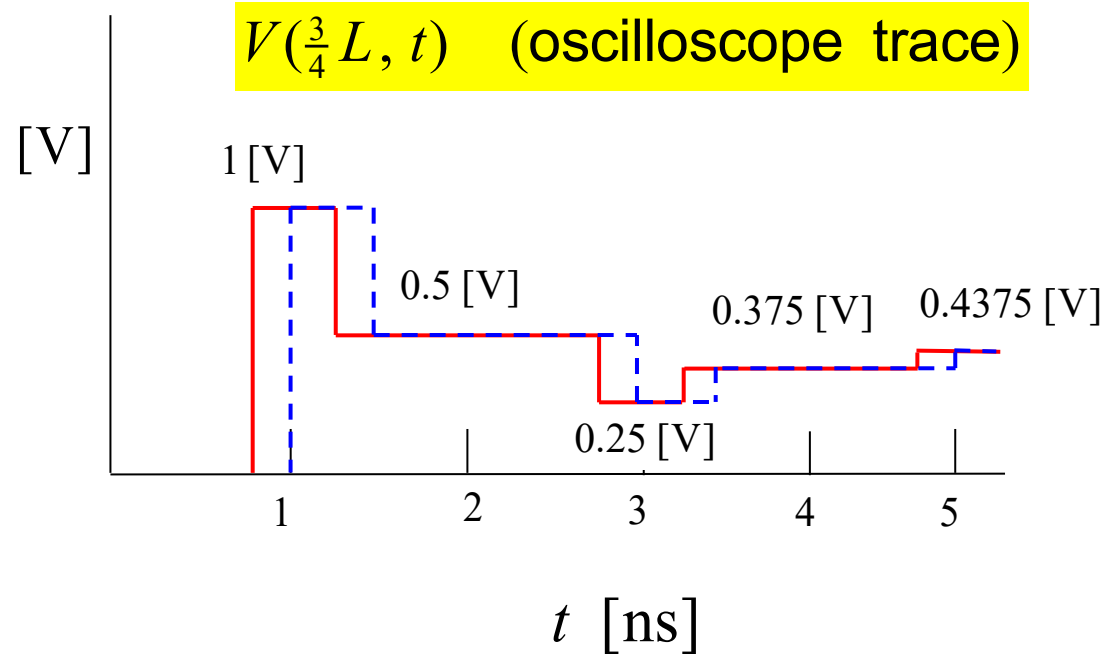


Pulse: Oscilloscope Trace



$$V_0 = 4 \text{ [V]}$$

$$W = 0.25 \text{ [ns]}$$



Example: Pulse (cont.)

